DigitalTwin.ai

Detailed Business Proposal · Accenture Innovation Challenge 2026, Problem Track 4 · Team HipHipHooray · Sagar Sahu, Priyansh Goyal · IIT Kharagpur

The one-line claim. A vehicle assembly line already emits enough data to say *which station is costing you cars right now*, *which one is about to*, and *what to do about it* — on a line where nearly a fifth of the stations have no sensors at all. We built that loop, ran it for **903 shifts**, and every number in this document is measured against held-out ground truth with its source named.

15.5%	0.025	70.5%	903
of forming-bottleneck warnings name a station with **no sensors at all** [15.0–15.9], n=29,060	calibration error (ECE) after fitting, from 0.479 600 held-out samples	of alarmed tools get an **actionable** diagnosis [67.9–72.9], n=1,246	shifts replayed through the live loop 86,742 decisions recorded

1 · Problem framing

A plant manager can already tell you last month's OEE. What no system reliably tells them is the question that actually decides output: **at 14:20 today, which single station is holding the line back, and is it worth walking to?**

Three things make that hard, and they are the three the brief names:

- **The constraint moves.** In our data it changes station roughly **20 times per shift**. A weekly report averages that away, and an average describes no moment of the shift it covers.
- **Utilisation lies.** A station that is blocked by its neighbour and a station that is genuinely slow both read ~95% busy. Ranking by busyness sends people to the victim.
- **The data is uneven.** Real lines mix legacy and modern equipment. Some stations are richly instrumented; others emit nothing at all and rely on a human with a clipboard.

The gap is not sensing and it is not dashboards. It is the step from *describing* the line to *deciding* on it — and doing so honestly enough that a supervisor keeps trusting the system after the third false alarm.

2 · Solution design

2.1 A twin, not a shadow

The distinction that matters (Kritzinger's taxonomy): a *model* is offline, a *shadow* receives data one-way, a *twin* closes the loop. Our prototype ingests, detects, ranks, prescribes and re-reads on a timer, and the causality is **enforced structurally**: the detector is handed a view of the run truncated at `now`, so it cannot read the future even by accident. Verified identical verdicts at 13/13 timepoints against the full-history run.

2.2 Four tiers of knowledge, never mixed

TIER	WHAT IT IS	WHERE IT COMES FROM
A — measured	The station reports it	PLC state, scans, tool telemetry
B — inferred	Bracketed between neighbours	Boundary timestamps, buffer slope
C — attested	A person says so	Manual checklist entry, with its entry latency
D — predicted	Projected under current flow	Buffer countdown

Every value on screen carries its tier. An operator must never see an inferred number and a measured one in the same font.

2.3 How the constraint is found

Rank by **effective cycle time** — work time divided by availability — not by busyness. Work time excludes seconds the station spent blocked or starved, so a victim is not charged for its neighbour's fault. Each station is compared to **its own early baseline**, never to takt: a station drifting 54→57 s is abnormal even while comfortably inside a 60 s takt.

2.4 From detection to prescription

The twin names the action and what ignoring it costs. The action vocabulary is a small library keyed on fault classes the detector itself separates; the **ranking and the cost are computed**, never retrieved. That is the step from descriptive → diagnostic → predictive → **prescriptive**.

> **The alert contract.** An alert may be raised only if it carries all five of: the ranked candidate and its margin; the evidence that produced it; a persistence estimate; the recommended action; and the expected cost of not acting. **An alert that cannot state its evidence is suppressed, not downgraded** — a quiet alert with no evidence is still noise, and noise is what erodes floor trust. In the recorded run, 7 alerts were suppressed by this rule.

3 · Target users

Three users, three resolutions, **one model** — and we prove they are one rather than asserting it.

USER	NEEDS	WHAT WE BUILT
Floor supervisor	Real time, in the moment	ISA-101 display: grey base, colour only on deviation. A well-designed plant HMI is visually quiet during normal operation. Dark stations are drawn hatched, not hidden.
Plant manager	Weekly planning	A **constraint-occupancy distribution**, deliberately *not* an average — the constraint moves ~20×/shift, so a mean describes no moment of it. "S20 held the line 21.3% of week 1" is a scheduling and capex input; a mean is not.
Leadership	Investment case	Value, coverage and the evidence table — every figure naming the file that produced it. Colour is allowed here: ISA-101 governs a plant HMI, not a business dashboard.

> **The reconciliation test.** Each level totals independently and is compared: **6,730 constraint-minutes and 4,431 vehicles, identical across supervisor records, manager weeks and leadership.** That is how we show three views are one twin instead of three dashboards.

4 · Business case and impact

4.1 Detection quality, measured against economic ground truth

Ground truth is not "which station looked busiest". It is the station whose speed-up **actually produces more cars**, measured by re-running the line under common random numbers. That makes the benchmark economic rather than definitional.

METHOD	TOP-1	TOP-2	REGRET (CARS/BLOCK)
Effective cycle time (ours)	43.1%	59.4%	**1.309**
Active period (Roser 2001)	46.0%	57.9%	1.348
Utilisation (the common baseline)	35.1%	48.0%	1.477

Top-1/top-2 on 202 strong-constraint blocks; regret on all 958 blocks. Source: `results/eval_v5.csv`.

What we may and may not claim. Paired McNemar tests on the same blocks: we **significantly beat the utilisation baseline ($p = 0.0025$)**, and we are **statistically tied with the active-period method ($p = 0.45$)**. We therefore claim the first and never the second. Comparing independent confidence intervals would have been the wrong test for same-block data and would have let us overclaim.

4.2 Why regret, and not accuracy

Two independent findings point the same way. First, the sensitivity labels carry a **0.79-car noise floor**, while the median margin between the best and second-best station is only 0.50 cars — so the top-1 label survives noise in only ~50% of blocks and is close to a coin flip. Second, across four line topologies **top-1 collapses from 44.5% to 10.6% while regret stays between 1.208 and 1.431**. The operationally meaningful metric transfers; the argmax metric does not.

4.3 Value, stated honestly

VALUE LINE	STATUS
Fewer cars lost to a mis-identified constraint (regret 1.309 vs 1.477 utilisation, on a 2.271-car ceiling)	**measured**
Earlier warning: buffer countdown, 178 warnings, median error +0.57 min	**measured**
Avoided false scrap: transducer-drift tools flagged for recalibration, not repair	**measured**
Sensor spend avoided: 15.5% of forming warnings need no sensor at that station	**measured**
Lead-time reduction under CONWIP release control	**NOT measured — excluded**

One value line has been deliberately removed. An earlier draft led the business case with "same throughput, 36% lower lead time, zero capital expenditure". No file in our results produces that number, so under our own rule — *a number is measured with a named source, cited with a named reference, or it does not appear* — it does not appear. We would rather present four defensible lines than five with one that dies under a single question.

5 · Handling the real-world complexities

#	COMPLEXITY	HOW WE ANSWER IT	STATE
1	Uneven sensor coverage; manual checklists	Dark stations are bracketed between neighbours and named directly by buffer slope — **15.5% of all forming warnings name a station with no sensors** [15.0–15.9], n=29,060. Manual checks enter as Tier C *attested* data carrying entry latency, because a human record does not exist for the twin until someone types it in.	Demonstrated (dark); schema built and tested for checklists
2	Multi-causal, intermittent root causes	Separated by **scope** — who else is affected and who is not — rather than by statistics. The prototype separates fault classes (slowing, breakdown, starved, blocked). **Operator variation is deliberately excluded on ethical grounds:** we measure the station, never the person. That is a design choice, stated, not an oversight.	Partly demonstrated; scope engine designed
3	PLC risk; retrofits only in maintenance windows	**The twin advises and never writes to line control** — a read-only boundary in ISA-95 terms, shown on every screen. Sensing additions mount externally and publish on a separate network, so no PLC program changes and therefore no re-validation.	Boundary demonstrated; window-dated schedule designed
4	Early defect surfaces late	Onset is read **backwards off the CUSUM accumulator**, then the VIN thread lists exactly which vehicles carry it, **partitioned by where they are** — a car on the line is a rework instruction, one that has shipped is a customer event. Onset dated to **+2 minutes** of truth.	Demonstrated
5	Three stakeholder views	One record stream, three resolutions, with the reconciliation test passing exactly.	Demonstrated
6	Scaling: layout, vintage, sensor maturity	Measured across four topologies including a parallel-server pair that breaks the pure-series assumption. See §6.	Demonstrated (layout, sensor maturity); vintage unmodelled
7	False alarms erode trust	Confidence is **calibrated, not asserted** (ECE 0.479→0.025), and every alert is logged with a human confirm/override so precision can be shown over time. **48.6 alerts/shift** against the ISA-18.2 budget of <150.	Demonstrated

6 · Does it survive a different line?

LINE	N BLOCKS	TOP-1	REGRET	CEILING
L1 — 20 stations, 1 merge *(reference)*	191	44.5%	1.208	4.272
L2 — 30 stations, 2 merges	94	10.6%	1.218	1.705
L3 — 20 stations + parallel pair	94	13.8%	1.431	1.870
L4 — 15 stations, 1 merge	96	21.9%	1.221	2.096

Source: `results/transfer_eval.csv`. L2–L4 are 12 runs each, so intervals are wide (L2 top-1 [5.9, 18.5]); these are directional, not definitive.

Sensor maturity. Running the entire pipeline from boundary scans with **zero PLC state tags** costs **+1.683 cars/block on L1** but only **+0.021 on L2**. State tags matter enormously where a strong constraint exists and almost not at all on a balanced line — so **instrument the lines with the most to gain first**. That is a more useful retrofit input than a single averaged degradation.

7 · Phased roadmap

PHASE	WHAT HAPPENS	GATE TO PASS BEFORE THE NEXT PHASE
0 · Shadow weeks 1–4	Subscribe to the streams the plant already produces and discards. No writes, no PLC changes.	Twin reproduces the plant's own throughput numbers
1 · One supervisor weeks 5–10	One line, one shift, one screen. Every alert confirmed or overridden by a person.	Running precision stable over ≥20 shifts; alerts within the ISA-18.2 budget
2 · Floor months 3–6	All shifts on that line. Manager view opens for planning.	Reconciliation holds; supervisors use it unprompted
3 · Retrofit next window	Costed sensor list installed at a scheduled shutdown, ranked by exposure closed per rupee.	Measured coverage improvement matches the prediction
4 · Second line months 6–12	Transfer. Report the commissioning curve honestly, not a yes/no.	Regret within the range measured across our four topologies
—	**Never: closed-loop write to line control.**	—

8 · Key risks and mitigations

RISK	WHY IT IS REAL	MITIGATION
False alarms erode trust	The brief names it; it is how these systems die	Calibrated confidence (ECE 0.025), the five-field contract with suppression, and a ledger showing running precision. 48.6 alerts/shift vs a 150 budget
Overconfidence in a weak signal	We have already been caught by this once	We measured our own overtake-risk mechanism at **5.9% correct [1.0–27.0], n=17** against 70–100% stated confidence, declared it failed, and removed it from the live path. The buffer countdown replaced it
Top-1 accuracy is a misleading target	0.79-car label-noise floor; argmax survives jitter in ~50% of blocks	Regret is the headline metric; top-1 reported with Wilson intervals alongside
Transfer may not hold	L2–L4 are 12 runs each	Stated as directional. Regret held 1.21–1.43 across four topologies; the commissioning curve is reported, not asserted
Touching production	Regulated, safety-certified control	Read-only by construction and enforced by a test, not a promise
Modelling people	Operator variation is a named cause and a real hazard	Excluded by choice. We measure the station, never the person, and say so

9 · What we deliberately did not do

- **Equipment vintage** is one of the three scaling axes the brief names and we have no data axis for it. Unmodelled, and stated.
- **Operator variation** — excluded on ethical grounds, as above.
- **The CONWIP lead-time figure** — removed for want of a source file.
- **Closed-loop control** — outside what any plant grants a prototype.

Every one of these is a decision we can defend, which is worth more than a claim we cannot.

All figures traceable to `results/` and `results/twin.db` (903 shifts, 86,742 recorded decisions). Reproduce with `scripts/query_twin.py`, `scripts/eval_v5.py`, `scripts/eval_transfer.py`, `scripts/fit_calibration.py`. Repository: github.com/sagar2907/HipHipHooray_IIT_Kharagpur_DigitalTwin.ai_Prototype